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EDUCATION

National Cheng Kung University

Master of Science in Computer Science and Information Engineering

Expected Sep 2025 – June 2027

Tainan, Taiwan

National Tsing Hua University

Bachelor of Science in Interdisciplinary Program of Science

Sep 2021 – Jun 2025

Hsinchu, Taiwan

COURSEWORK / SKILLS

Core CS Foundations

- Operating Systems
- Data Structures & Algorithms
- Computer Networks
- Database Systems
- Computer Architecture

Software, Systems & Tech

- Software Development
- Network Security
- Blockchain Technologies
- High Performance Computing
- Competitive Programming

AI & Data Science

- Artificial Intelligence
- Machine Learning
- Data Science
- Statistics
- Time Series Analysis

PROJECTS

Undergraduate Research: Network Attack Detection | Python, ML

12.2023 - 10.2024

- Built the end-to-end data pipeline: feature standardization, **SMOTE** for class imbalance, and a unified train/val/test split framework to ensure fair model comparison.
- Team compared **GA+KNN** vs. **Boosting** (LightGBM 93.44% accuracy); clean data and rigorous evaluation proved more impactful than model complexity.

Soul Warrior (2D Game) | TypeScript, Cocos Creator, Firebase

02.2023 - 06.2023

- Implemented player movement, **combat mechanics** (melee hit-detection synced with animations, ranged attacks with **object-pooling**), and damage-feedback effects (invincibility frames, screen shake).
- Participated in **Firebase** integration for email authentication and cloud save/load with async data handling.

Bomberman (2D Game) | C++, Allegro

09.2022 - 01.2023

- Engineered a multiplayer game using **C++** and Allegro 5, implementing an **Object-Oriented architecture**.
- Designed an NPC behavior system using **BFS** per-frame distance maps to dynamically switch among **pursuit, evasion, and patrol** states based on player and danger-zone proximity.
- Manually optimized memory management to ensure stability during intensive local multiplayer matches.

COMPETITIONS

Agent for Truth: Disinformation Defense Hackathon | Full-Stack & AWS

03.2026

- Built the entire product around teammates' **LightGBM** model: **FastAPI** backend with SHAP explainability and **Amazon Bedrock Claude** risk summaries.
- Developed a **React + D3.js** risk-control dashboard featuring an interactive transaction graph with drag, zoom, and click-to-navigate.
- Designed **AWS** architecture (**CloudFront + S3** frontend, **ALB + ECS Fargate** backend) with Docker for deployment consistency.

Meichu Hackathon | Firmware Developer

10.2024

- Developed **touchscreen UI and interaction logic** on an NXP i.MX RT1060 embedded board for an air-quality monitoring IoT prototype in a 3-person team.
- Team pivoted to a **CSV static-data strategy** when embedded API integration proved infeasible within the 2-day timeframe, delivering a stable demo.

APAC HPC-AI Competition (Runner-up) | Team Member

06.2022 - 11.2022

- Achieved the **Runner-up prize** by optimizing *Quantum Espresso* simulation on the Gadi supercomputer.
- Profiled workload to identify **20 sub-problems**, then tuned MPI/CPU partitioning across single-node and multi-node configs; best run **~10s on 25 nodes / 1200 cores**, identifying the communication-overhead saturation point.

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, TypeScript, SQL, R, Go

Developer Tools: VS Code, Vim

Frameworks & Runtimes: ReactJS, NodeJS, FastAPI, PyTorch, Scikit-Learn, Pandas, Cocos Creator, Allegro

Platforms & DevOps Tools: AWS, Docker, Firebase, Linux, Git, Conda